education sciences

MDPI

Article

# Modeling Teachers' Acceptance of Generative Artificial Intelligence Use in Higher Education: The Role of AI Literacy, Intelligent TPACK, and Perceived Trust

**Ahlam Mohammed Al-Abdullatif**

Curriculum and Instruction Department, College of Education, King Faisal University, Eastern Province, Al-Ahsa, P.O. Box 400, Hofuf 31982, Saudi Arabia; aalabdullateef@kfu.edu.sa

**Abstract:** This study delves into the factors that drive teachers' adoption of generative artificial intelligence (GenAI) technologies in higher education. Anchored by the technology acceptance model (TAM), the research expands its inquiry by integrating the constructs of intelligent technological pedagogical content knowledge (TPACK), AI literacy, and perceived trust. Data were gathered from a sample of 237 university teachers through a structured questionnaire. The study employed structural equation modeling (SEM) to determine the relationships among the constructs. The results revealed that both AI literacy and perceived ease were the most influential factors affecting teachers' acceptance of GenAI. Notably, intelligent TPACK and perceived trust were found to be pivotal mediators in this relationship. The findings underscore the importance of fostering AI literacy and adapting intelligent TPACK frameworks to better equip educators in the age of AI. Furthermore, there is a clear need for targeted professional development initiatives focusing on practical training that enhances AI literacy. These programs should provide hands-on experience with GenAI tools, boosting educators' confidence and ability to integrate them into their teaching practices.

**Keywords:** generative artificial intelligence; GenAI; AI literacy; intelligent TPACK; acceptance; teachers

check for updates

**Citation:** Al-Abdullatif, A.M. Modeling Teachers' Acceptance of Generative Artificial Intelligence Use in Higher Education: The Role of AI Literacy, Intelligent TPACK, and Perceived Trust. *Educ. Sci.* **2024**, *14*, 1209. https://doi.org/10.3390/educsci14111209

Received: 2 October 2024
Revised: 23 October 2024
Accepted: 1 November 2024
Published: 3 November 2024

**Copyright:** © 2024 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

## 1. Introduction

The landscape of modern education is in continuous flux, punctuated by the emergence of cutting-edge technologies that redefine teaching and learning paradigms. Among these technologies, generative artificial intelligence (GenAI) technologies, with their profound capacity to simulate content generation and learning dynamics, have recently emerged at the forefront of educational technology discussions, offering tools and methodologies that promise to revolutionize educational processes [1]. However, similarly to other educational tools, GenAI's successful implementation in classrooms hinges on educators' willingness to embrace and integrate it into their teaching and learning strategies. Several interconnected factors influence this active adoption. While Davis's [2] technology acceptance model (TAM) offers valuable insights into educators' adoption of new tools, focusing solely on perceived ease of use and usefulness may be insufficient. GenAI's transformative potential presents a unique challenge, and understanding its acceptance among educators necessitates an exploration beyond traditional models. The introduction of intelligent technological pedagogical content knowledge (TPACK) offers a lens through which to view this acceptance, aiming to combine technological competence with pedagogical expertise specific to AI [3]. Similarly, AI literacy emerges as a critical determinant, underscoring the importance of educators being well versed with the principles, ethical implications, and nuances of AI technologies [4]. However, knowledge and understanding alone may not suffice. Perceived trust in GenAI plays a potentially pivotal role in this acceptance narrative [5]. Trust in this context not only pertains to the reliability and accuracy of the tool but also encapsulates broader concerns about security, ethical implications, and long-term impacts on the educational landscape.

While studies have investigated various dimensions of technological integration in classrooms, the specific drivers shaping teachers' intentions to adopt GenAI remain underexplored, necessitating the present research. A recent systematic review by Crompton and Burke [6] found a significant gap in the research on AI adoption in higher education. While 71% of studies focused on students, only 17% explored the perspectives of university teachers. This imbalance is concerning, as many studies have stressed the crucial role teachers play in introducing GenAI tools into classroom practice [7,8]. More research is needed to understand how teachers perceive and approach GenAI adoption [9,10]. Therefore, this study delves into the drivers that influence teachers' intentions to use GenAI, casting a spotlight on the interplay between technology acceptance, intelligent TPACK, AI literacy, and perceived trust. This study endeavors to model the structural relationships among these determinants, aiming to provide a comprehensive framework that elucidates the intricate dynamics influencing teachers' intentions towards GenAI. In doing so, we hope to offer stakeholders—educators and policymakers—insights that can guide the seamless and effective integration of GenAI into our classrooms.

## 2. GenAI Technologies: An Overview

GenAI, primarily characterized by its ability to generate new content through understanding and learning from existing data, represents a significant advancement in AI technology [11]. At the core of GenAI are sophisticated algorithms capable of performing tasks that traditionally require human intelligence. These technologies, as detailed by Zhang and Aslan [12], are not only transforming the way content is created but are also revolutionizing interaction within digital environments. Central to this revolution are large language models (LLMs), such as GPT-3 and GPT-4, which, as discussed by Bozkurt [7], are at the forefront of GenAI and provide the backbone for various educational applications. These models process and generate language-based outputs that can be used for creating learning content, automating feedback, and even interacting with students through conversational agents [1]. These systems utilize user input and learning data to dynamically produce materials that are tailored to the curriculum and the learner's pace, making education more adaptive and engaging [13]. GenAI also extends into predictive analytics, where it helps educators and administrators anticipate student needs, predict performance, and tailor interventions [9]. These technologies are instrumental in developing applications that range from automated content generation to intelligent tutoring systems, and offer scalable solutions that can be adapted across diverse educational settings [3,14].

## 3. GenAI in Education: Potential and Prospects

The integration of GenAI into educational systems offers several key benefits that can significantly enhance both teaching and learning experiences. One of the foremost benefits is the personalization of learning. GenAI enables the creation of adaptive learning systems that tailor educational content to meet the unique needs of each student, thereby accommodating various learning styles and paces [11,15,16]. This personalization is crucial in addressing the educational disparities that exist due to different learning capabilities [8,17]. This level of customization ensures that learning is more engaging and effective, addressing the specific educational needs of students and potentially reducing dropout rates. For example, AI-driven platforms can dynamically adjust the complexity of tasks based on a student's performance, ensuring that each learner is adequately challenged but not overwhelmed [7].

Moreover, GenAI facilitates a more interactive and engaging learning environment. Tools such as AI-powered tutors and virtual labs can provide students with hands-on learning experiences that are both engaging and informative [13]. These tools also support the continuous assessment and feedback mechanisms that are vital for effective learning [18,19]. The application of GenAI in administrative and instructional processes leads to significant efficiencies. AI systems can automate routine tasks, such as grading, scheduling, and student feedback, freeing educators to focus more on interactive and higher-level teaching

activities [8]. Additionally, AI-driven analytics can help educational institutions better manage resources by predicting enrolment patterns, optimizing class sizes, and identifying areas where interventions are most needed [14,16,20].

Despite the promising prospects of GenAI in education, there are significant challenges that must be addressed to harness its full potential. The deployment of GenAI raises substantial ethical questions, particularly concerning data privacy and security. The use of sensitive student data to train AI models poses risks related to data breaches and misuse [21,22]. Moreover, there is a need to establish clear guidelines to ensure that AI systems do not perpetuate biases or make unfair decisions, particularly in assessments and admission processes [14,22]. The use of AI in educational settings must be regulated to protect students' privacy and ensure that the systems do not perpetuate existing biases or create new ones [23,24]. Another major challenge is overreliance on AI for educational purposes, which could lead to the diminishment of traditional educational skills among students, such as critical thinking and problem-solving [18,25,26].

## 4. Theoretical Foundation and Hypothesis Formulation

### *4.1. TAM Constructs*

TAM, formulated by Davis [2], has become a widely referenced framework for assessing how users decide to accept or reject technologies. According to Davis [27], the two fundamental elements of TAM—perceived usefulness and perceived ease of use—greatly impact an individual's decision to embrace or discard a technology. The belief that technology will improve job performance is known as perceived usefulness, while perceived ease of use suggests that it is user-friendly and effortless to operate. According to TAM, the relationship between these two concepts is such that perceived ease of use positively influences perceived usefulness. This means that if a technology is easy to use, users are more likely to perceive it as useful. This is because technologies that are straightforward and simple to operate minimize the effort required to achieve the desired outcomes, thereby enhancing the user's performance or the perceived benefits of using the technology. The higher these perceptions, the more favorable the user's attitude towards the technology, thereby increasing the likelihood of its adoption in future scenarios.

In the realm of education, and more specifically in the context of teachers' adoption of novel technologies, TAM provides a robust theoretical lens. With the surge in AI-driven tools and applications within educational settings, understanding educators' intention to incorporate AI has become paramount. Several studies have applied TAM to investigate teachers' perceptions and adoption of AI technologies. Chen et al. [11] and Liao et al. [28] used TAM to assess educators' acceptance of AI-driven educational tools, emphasizing the role of perceived usefulness in shaping positive attitudes and perceptions. Similarly, Malik et al. [29] highlighted the significance of perceived ease of use in influencing students' intentions to integrate AI chatbots into their teaching methodologies. Another study, by Kaplan-Rakowski et al. [9], indicated that perceived use had the strongest influence on teachers' perceptions of adopting GenAI in teaching and learning practices. Drawing on the TAM model and considering the previous discussion, we formulated the following hypotheses (Hs):

**H1:** *Perceived usefulness affects the acceptance of GenAI.*

**H2:** *Perceived ease of use affects the acceptance of GenAI.*

**H3:** *Perceived ease of use affects the perceived usefulness of GenAI.*

### *4.2. Extended TAM*

Despite TAM's widespread application, the model continues to be refined and expanded, incorporating new variables and adapting to contemporary technological advances and cultural shifts. A recent systematic review by Rosli et al. [30] on the advancements

in and adaptations of TAM in the context of higher education highlighted that despite the robust framework of TAM, there are emerging gaps, particularly in understanding how these constructs interact with newer technologies, such as the GenAI in this study. This points to the need for more nuanced studies that also consider aspects related to GenAI to capture the full spectrum of its acceptance and usage. In the context of this study, extending TAM with factors such as intelligent TPACK, AI literacy, and perceived trust is pivotal in adapting to the evolving landscape of GenAI technologies in higher education. Intelligent TPACK ensures that university teachers are equipped to integrate AI-based tools within their specific pedagogical contexts, enhancing the relevance and effectiveness of GenAI in teaching and learning [3]. AI literacy is crucial because it empowers teachers to understand and interact confidently with AI systems, thus addressing the skills gap that might hinder effective utilization [31]. Meanwhile, perceived trust addresses the significant security-related barriers and ethical concerns that teachers may have with AI [5]. Together, these extended factors contribute to a more comprehensive understanding of GenAI acceptance, promoting smoother integration and more effective use of GenAI technologies across various fields in higher education. Incorporating these factors into TAM can provide deeper insights into specific challenges and requirements for the widespread adoption of GenAI technologies. The following section introduces the extended TAM factors of intelligent TPACK, AI literacy, and perceived trust.

### *4.3. Intelligent TPACK*

The concept of TPACK has evolved significantly since Shulman introduced pedagogical content knowledge, which Mishra and Koehler developed later to include technological knowledge [32]. This framework, now crucial in integrating technology into education, has been further adapted to accommodate the use of AI, leading to the development of intelligent TPACK. Intelligent TPACK measures teachers' proficiency in selecting and using AI-based tools effectively within educational settings [3]. While TPACK seeks to equip educators with an understanding of how technology can be seamlessly embedded into teaching [33,34], intelligent TPACK dives deeper, advocating the integration of AI-driven tools and platforms. This involves not just the use of AI tools but also an understanding of AI's capabilities, potential pitfalls, ethical considerations, and overarching influence on pedagogical strategies and content delivery [35]. By considering intelligent TPACK, teachers can align AI technologies with curriculum needs and pedagogical strategies, ensuring that technology integration is meaningful and enhances learning outcomes [3].

TAM posits that two primary factors influence an individual's intention to use a technology: perceived ease of use and perceived usefulness [27]. Teachers' perceived ease of use of GenAI is significantly influenced by their understanding of intelligent TPACK. An educator experienced in intelligent TPACK will likely find GenAI tools more accessible and user-friendly. A strong foundation in this framework equips teachers with the skills to implement AI tools that personalize learning and enhance educational outcomes [36]. The perceived usefulness of GenAI tools is closely tied to the extent of intelligent TPACK among educators. This knowledge reduces the perceived complexity of GenAI technologies. With a grasp of AI's potential and its integration with teaching strategies, educators can discern the tangible benefits of AI-based tools in enhancing learning experiences and student engagement and performance, thereby perceiving them as more useful [37]. When perceived ease of use and perceived usefulness converge, they have a significant impact on an educator's intention to use GenAI technologies. Thus, by fostering a strong foundation in intelligent TPACK, we can enhance teachers' willingness and confidence in adopting GenAI in their teaching practices.

Studies have shown that robust TPACK facilitates the adoption and integration of new technologies into teaching practices [38]. Studies such as Hsu [39] and Joo et al. [33] have found that perceived ease of use, perceived usefulness, and technology acceptance can be predicted by teachers' TPACK in educational technology adoption. Likewise, Yang et al. [40] highlighted the positive impact of teachers' TPACK on their perceived ease

of use, usefulness, and overall acceptance of E-Schoolbag. Similar results were supported by other studies, such as Mailizar et al. [41] in the context of online perfectional development acceptance, and Jang et al. [42] and Thohir et al. [43] regarding augmented and virtual reality acceptance. Given the growing body of literature supporting the significant role of TPACK in predicting teachers' technology acceptance, the following hypotheses are proposed:

**H4:** *Intelligent TPACK affects the perceived usefulness of GenAI.*

**H5:** *Intelligent TPACK affects the perceived ease of use of GenAI.*

**H6:** *Intelligent TPACK affects the acceptance of GenAI.*

### *4.4. AI Literacy*

AI literacy is an evolving field focused on defining, teaching, and assessing knowledge and skills associated with AI technology [31]. It encompasses an individual's capability to understand, apply, monitor, and critically assess AI applications, even without the technical expertise to create AI systems [4,31,44,45]. A comprehensive definition by Long and Magerko [46] describes AI literacy as "a set of competencies that enables individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool online, at home, and in the workplace" (p. 2). This approach to AI literacy equips individuals to adeptly navigate and contribute to a world increasingly dominated by AI technologies.

In educational contexts, the integration of AI into teaching and learning has intensified the focus on AI literacy [47]. It is crucial for students to develop the ability to judiciously utilize AI technologies and to distinguish between ethical and unethical uses [22,37]. An individual's literacy skills significantly influence their execution of learning activities [42,48]. Numerous studies have underscored the benefits of high AI literacy on interactions between humans and AI. For instance, Su and Yang [49] observed that AI literacy programs enhance students' knowledge, skills, and attitudes towards AI, suggesting that such literacy contributes positively to students' perceptions of and competence using AI technologies. According to Wang et al. [47], the degree of AI literacy can determine individuals' readiness to embrace AI technologies. Educators with higher AI literacy are more likely to appreciate the potential enhancements that AI can bring to educational environments, thus fostering a more receptive attitude towards accepting these technologies [48].

When applied to GenAI, AI literacy equips users with a more profound understanding of both the potentials and constraints of AI technology, fostering more effective implementation and usage [46]. This knowledge inherently reduces hesitations or misconceptions that might arise from unfamiliarity [31,47]. Those well versed in AI are more skilled at identifying the AI tools and solutions that best fit their needs [4]. This proficiency allows them to critically assess various AI products, enhancing their appreciation of the advantages these technologies offer. Lee et al. [50] investigated the impact of information literacy on user trust in government websites, revealing that higher levels of perceived information literacy improved perceptions of the websites' usefulness. Teachers who understand AI concepts and applications are more likely to recognize the benefits of GenAI, such as personalized learning and efficient student assessment, and therefore perceive these technologies as more useful [45,51]. Therefore, the following hypothesis is proposed:

**H7:** *AI literacy affects the perceived usefulness of GenAI.*

An AI-literate educator familiar with the intricacies of AI tools is more likely to find these tools user-friendly and accessible. This familiarity reduces complexity and increases the comfort level when using AI tools [11,52]. Studies have shown that educators who are more literate in AI find it easier to adopt and interact with AI technologies because they are better able to understand and navigate the functionalities of these systems [53]. As a result, the following hypothesis is proposed:

**H8:** *AI literacy affects the perceived ease of use of GenAI.*

While intelligent TPACK offers a framework for educators to integrate AI tools effectively in association with pedagogy and content [3], AI literacy provides the foundational knowledge to make this integration informed and strategic [4]. AI literacy enriches the technological domain of intelligent TPACK. An educator grounded in AI principles can better discern which tools align with specific pedagogical goals, ensuring more seamless and impactful integration [48]. With AI literacy, the application of AI tools becomes more purpose-driven within the intelligent TPACK framework [35]. Educators can move beyond mere tool adoption to tool optimization, extracting maximum pedagogical value from AI integrations. Consequently, the following hypothesis is proposed:

**H9:** *AI literacy affects teacher's intelligent TPACK.*

AI literacy is a strong predictor of teachers' acceptance of GenAI. Higher levels of AI literacy facilitate a greater understanding of the advantages and capabilities of AI, which in turn promotes a positive attitude towards its acceptance and integration in educational settings [4,31,47]. This acceptance is driven by the alignment of AI capabilities with pedagogical goals and the potential for improved educational outcomes [54]. Hence, the following hypothesis is proposed:

**H10:** *AI literacy affects the acceptance of GenAI.*

### *4.5. Perceived Trust*

Perceived trust can be defined as the subjective belief or confidence that a user has in the reliability, integrity, and safety of a system or technology [55]. In the context of GenAI, it pertains to a user's convictions about the accuracy, ethical operations, security, and dependability of AI-based tools and applications in educational settings [56]. Generative AI by design can create content, resources, and scenarios. The lack of trust stemming from an insufficient understanding of AI operations contributes to resistance among teachers towards the integration of AI-based tools into their instructional practices [5,56]. For educators, trusting that this generated material is accurate, relevant, and free from bias is essential [57]. Their trust mitigates the perceived risks associated with using AI-driven tools in classrooms [58]. Trust acts as a catalyst: when educators believe in the integrity and reliability of GenAI tools, they are more likely to adopt and integrate them into their pedagogical strategies. An inherent trust in GenAI ensures that teachers feel confident about the quality and appropriateness of AI-generated content, ensuring that they can use these resources confidently without constant doubt or the need for verification.

Trust in GenAI is closely linked to AI literacy. Teachers with a comprehensive understanding of AI are more likely to trust the technology's reliability and effectiveness in educational settings [59]. This trust stems from a deeper comprehension of how AI works and its potential impact on teaching and learning outcomes [47]. A deeper understanding of AI operations and their implications helps build trust in these technologies, making educators more confident in their deployment and expected benefits [46,47]. The relationship between AI literacy and trust in GenAI is further supported by studies showing that higher levels of AI literacy correlate with more profound trust in and acceptance of AI applications in educational settings. Lee et al. [50] indicated that individuals' information literacy has a significant influence on their perceived trust in using government websites. Similarly, a study by Nazzal et al. [60] confirmed that digital literacy is a strong predictor of user trust in the e-commerce context. This results in the proposal of the following hypothesis:

**H11:** *AI literacy affects the perceived trust in GenAI.*

The relationship between perceived trust in and perceived usefulness of GenAI is underscored by findings that trust in GenAI's capabilities directly influences its perceived

effectiveness in educational settings. If a tool is not perceived as trustworthy, its utility is immediately questioned, regardless of its potential capabilities [56]. On the other hand, a trusted GenAI tool lays the groundwork for its content or outputs to be deemed useful. When teachers trust that the GenAI tool consistently generates accurate and relevant content, they inherently perceive the tool as more useful. Many studies have demonstrated that when educators trust AI-based tools, they are more likely to recognize and utilize these tools' capabilities in enhancing teaching and learning outcomes [5,58]. This leads to the following hypothesis:

**H12:** *Perceived trust affects the perceived usefulness of GenAI.*

## 5. Materials and Methods

### 5.1. Sample and Data Collection

During April and May of the 2023 academic year, a survey was conducted among [Redacted] University teachers after securing ethical clearance and informed consent, as per the university's ethics committee guidelines, under reference number [Redacted]. The inclusion criteria for this study required that participants be university teachers, including lecturers, assistant professors, and professors, who were actively engaged in teaching. Additionally, participants needed to have had at least some prior encounter with generative AI, either through its use in their teaching practices, research, or exploration of AI tools in educational contexts. Only those who voluntarily consented to participate in the study were included. The survey was disseminated through university email systems and various social media channels. Participants were allowed a four-week period to voluntarily respond to the survey. We invited approximately 600 university teachers and received 237 completed responses, yielding a response rate of 40%. The sample size of 237 was adequate for SEM analysis, as Kline [61] recommends a minimum of 200 participants, particularly for models with multiple constructs. This ensures the analysis is statistically reliable and valid. A detailed profile of the sample is presented in Table 1. Of those who responded, 77.6% identified as female, with a majority being lecturers and assistant professors (74.6%) aged between 25 and 44 years (72.1%). Most had experience ranging from less than 5 years (27.8%) to 5–10 years (26.6%) and 11–20 years (29.1%). The participants represented various academic fields, as shown in Table 1.

**Table 1.** Sample profile (n = 237).

| Characteristic | n | % |
|---|---|---|
| Gender | | |
| Male | 53 | 22.4 |
| Female | 184 | 77.6 |
| Age | | |
| <25 | 6 | 2.5 |
| 25–34 | 84 | 35.4 |
| 35–44 | 87 | 36.7 |
| 45–54 | 45 | 19.0 |
| >54 | 15 | 6.3 |
| Teaching experience, years | | |
| <5 | 66 | 27.8 |
| 5–10 | 63 | 26.6 |
| 11–20 | 69 | 29.1 |
| 21–30 | 33 | 13.9 |
| >30 | 6 | 2.5 |

**Table 1.** *Cont.*

| Characteristic | n | % |
|---|---|---|
| Academic rank | | |
| Teaching assistant [1] | 21 | 8.9 |
| Lecturer [2] | 84 | 35.4 |
| Assistant professor [3] | 93 | 39.2 |
| Associate professor | 27 | 11.4 |
| Full professor | 12 | 5.1 |
| Academic major | | |
| Health | 101 | 42.6 |
| Humanities and social sciences | 66 | 27.8 |
| Agriculture | 29 | 12.2 |
| Engineering | 3 | 1.3 |
| Natural sciences | 12 | 5.1 |
| Computer science and information technology | 26 | 11.0 |

[1] Teaching assistant: holds a bachelor's or higher diploma; [2] Lecturer: holds a master's degree, [3] Assistant professor: holds a PhD.

### 5.2. Data Analysis and Measurement

The analysis was conducted using the partial least squares structural equation modeling (PLS-SEM) method, facilitated by SmartPLS 4.0 software. As outlined by Hair et al. [62], this method entails a two-stage process: the first analyzing the measurement model and the second examining the structural model to test the research hypotheses linked to the constructs. The study examined participants' views on six constructs of the research model (illustrated in Figure 1) using a previously developed survey containing 31 questions (see Appendix A). The data collection process commenced with demographic data, followed by the evaluation of perceptions using a five-point Likert scale. The six constructs evaluated included perceived usefulness (four items); perceived ease of use (four items); GenAI acceptance (four items), adopted from Davis [27]; intelligent TPACK (six items), adopted from Celik [3]; AI literacy (four items), adopted from Wang et al. [47]; and perceived trust (five items), adopted from Nazaretsky et al. [59].

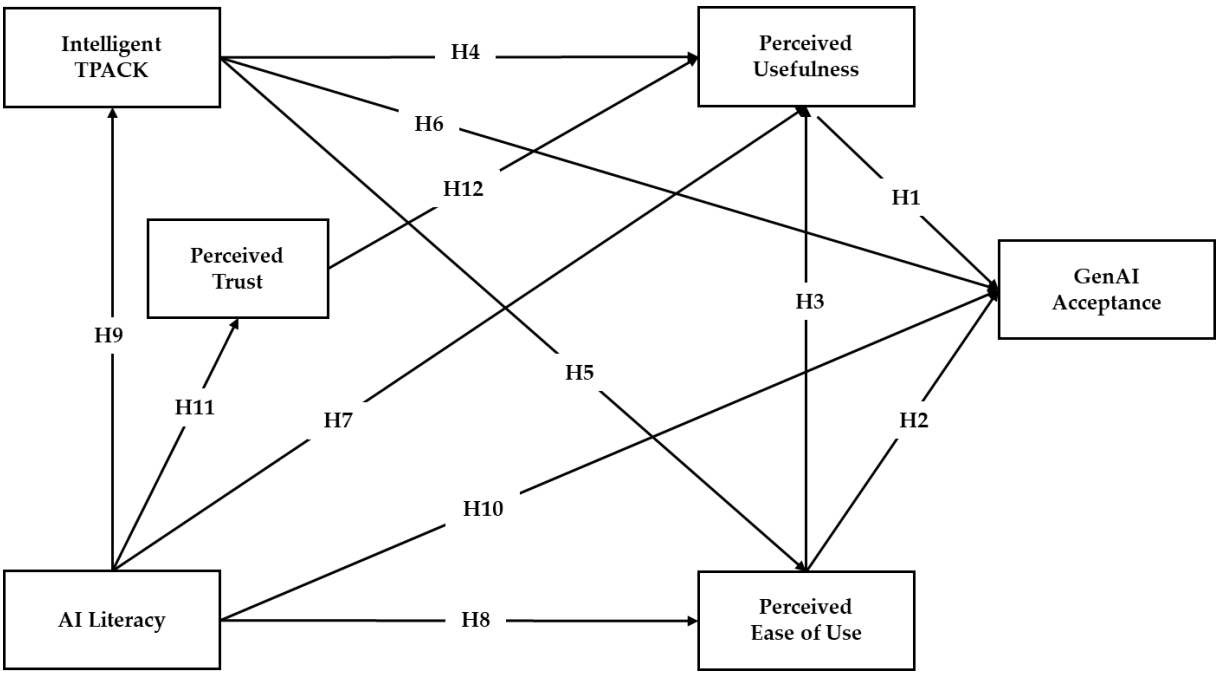


**Figure 1.** Proposed research model.

## 6. Results

### 6.1. Measurement Model Analysis

The process began with validation of the construct validity to ensure an accurate representation of the intended concepts [62]. Table 2 illustrates the indicator loading. All values are between 0.72 and 0.94, suggesting satisfactory to high levels of loading, as recommended by Hair et al. [62]. Next, the internal consistency was confirmed through Cronbach's alpha ($\alpha$) and composite reliability (CR), with values of $0.81 \geq \alpha \leq 0.94$ and $0.87 \geq CR \leq 0.95$ indicating good to high reliability. As recommended by Hair et al. [62], convergent validity was affirmed through the average variance extracted (AVE) for each construct (AVE > 0.5).

**Table 2.** Construct loading, convergent validity, and reliability analysis.

| Construct | Indicator | Indicator Loading | $\alpha$ | CR | AVE | $R^2$ | $R^2$ Adjusted | $Q^2$ |
|---|---|---|---|---|---|---|---|---|
| Perceived ease of use | Item 1 | 0.72 | 0.81 | 0.87 | 0.63 | 0.29 | 0.27 | 0.10 |
| | Item 2 | 0.83 | | | | | | |
| | Item 3 | 0.82 | | | | | | |
| | Item 4 | 0.79 | | | | | | |
| Perceived usefulness | Item 1 | 0.86 | 0.86 | 0.90 | 0.70 | 0.38 | 0.34 | 0.02 |
| | Item 2 | 0.82 | | | | | | |
| | Item 3 | 0.80 | | | | | | |
| | Item 4 | 0.86 | | | | | | |
| Perceived trust | Item 1 | 0.82 | 0.91 | 0.93 | 0.72 | 0.10 | 0.08 | 0.05 |
| | Item 2 | 0.92 | | | | | | |
| | Item 3 | 0.75 | | | | | | |
| | Item 4 | 0.87 | | | | | | |
| | Item 5 | 089 | | | | | | |
| AI literacy | Item 1 | 0.91 | 0.92 | 0.94 | 0.79 | | | |
| | Item 2 | 0.92 | | | | | | |
| | Item 3 | 0.87 | | | | | | |
| | Item 4 | 0.86 | | | | | | |
| Intelligent TPACK | Item 1 | 0.89 | 0.94 | 0.95 | 0.77 | 0.42 | 0.41 | 0.40 |
| | Item 2 | 0.83 | | | | | | |
| | Item 3 | 0.89 | | | | | | |
| | Item 4 | 0.93 | | | | | | |
| | Item 5 | 0.91 | | | | | | |
| | Item 6 | 0.80 | | | | | | |
| GenAI acceptance | Item 1 | 0.88 | 0.91 | 0.93 | 0.78 | 0.65 | 0.63 | 0.38 |
| | Item 2 | 0.94 | | | | | | |
| | Item 3 | 0.92 | | | | | | |
| | Item 4 | 0.79 | | | | | | |

Discriminant validity was established through the heterotrait–monotrait (HTMT) ratio, ensuring distinctiveness among constructs. Ideally, this ratio should be greater than 0.85 to indicate discriminant validity. The square roots of the AVE values for all constructs (illustrated in Table 3) exceeded their respective intercorrelations, with HTMT values consistently below 0.85.

**Table 3.** Correlation matrix and discriminant validity analysis.

| Constructs | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1. Perceived ese of use | **0.79** | | | | | |
| 2. Perceived usefulness | 0.44 (0.53) | **0.84** | | | | |

**Table 3.** *Cont.*

| Constructs | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 3. Perceived trust | 0.24 (0.31) | 0.48 (0.51) | **0.85** | | | |
| 4. AI literacy | 0.39 (0.42) | 0.25 (0.27) | 0.30 (0.29) | **0.89** | | |
| 5. Intelligent TPACK | 0.53 (0.58) | 0.40 (0.44) | 0.27 (0.28) | 0.30 (0.68) | **0.88** | |
| 6. GenAI acceptance | 0.69 (0.79) | 0.47 (0.52) | 0.34 (0.37) | 0.60 (0.64) | 0.58 (0.62) | **0.88** |

Values in bold are the root squares of the AVE; values in parentheses are the HTMT ratios.

### *6.2. Structural Model Analysis*

The focus then shifted to the structural model in which the relationship between constructs was analyzed. To this end, Hair et al. [62] suggested presenting values such as standardized path coefficients (β), standard errors (SE), t-values (t), and *p*-values. The results are summarized in Figure 2 and Table 4. Perceived ease of use (H2: β = 0.48, t = 4.53**) and AI literacy (H10: β = 0.35, t = 3.83**) were found to be the two factors directly affecting GenAI's acceptance, which means that both H2 and H10 were confirmed. However, perceived usefulness (H1: β = 0.16, t = 1.35) and intelligent TPACK (H6: β = 0.04, t = 0.40) showed no direct effect on teachers' acceptance of GenAI. Thus, H1 and H6 were rejected. Furthermore, the results indicated that AI literacy was directly influenced by both intelligent TPACK (H9: β = 0.64, t = 8.55**) and perceived trust (H11: β = 0.30, t = 2.44*). Accordingly, H9 and H11 were supported. However, AI literacy had no direct effect on either perceived usefulness (H7: β = −0.13, t = 0.91) or perceived ease of use (H8: β = 0.08, t = 0.49), which resulted in the rejection of H7 and H8. In addition, the results showed that perceived usefulness was directly affected by both perceived ease of use (H3: β = 0.28, t = 2.23*) and perceived trust (H12: β = 0.39, t = 3.38**), supporting H3 and H12. In regard to teachers' intelligent TPACK, the result showed a significant direct effect on perceived ease of use (H5: β = 0.49, t = 3.02**) and an insignificant direct effect on perceived usefulness (H4: β = 0.23, t = 1.47), resulting in the acceptance of H5 and rejection of H4.

Although the results in Table 4 reveal an insignificant direct effect of intelligent TPACK on GenAI acceptance (H6: β = 0.04, t = 0.40), they show a significant indirect effect of intelligent TPACK on GenAI acceptance through the mediation of perceived ease of use (H6: β = 0.23, t = 2.36*). The total effect of intelligent TPACK on GenAI was significant (H6: β = 0.33, t = 3.20**). This means that perceived ease of use fully mediates the relationship between intelligent TPACK and GenAI acceptance. That is, intelligent TPACK first impacts perceived ease of use, which in turn influences GenAI acceptance. Similarly, the data presented in Table 4 indicate the non-significant direct influence of AI literacy on perceived usefulness (H7: β = −0.13, t = 0.91). However, AI literacy was found to have a direct impact on perceived usefulness, facilitated by the mediating role of perceived trust (H7: β = 0.12, t = 2.10*). The total effect of AI literacy on perceived usefulness was significant (H7: β = 0.25, t = 2.01*). This suggests that the relationship between AI literacy and perceived usefulness is entirely mediated by perceived trust. In addition, the data in Table 4 disclose that AI literacy's direct impact on perceived ease of use was not statistically significant (H8: β = 0.08, t = 0.49). However, there was a significant indirect influence of AI literacy on perceived ease of use, facilitated through the intermediary role of intelligent TPACK (H8: β = 0.31, t = 2.54*). The total effect of AI literacy on perceived ease of use proved to be statistically significant (H8: β = 0.39, t = 3.49**). This implies that the link between AI literacy and perceived ease of use is fully mediated by intelligent TPACK.

The $R^2$ values indicated a high level of predictive capability for the key outcomes, with the $Q^2$ values supporting the model's predictive relevance for out-of-sample validation [62,63]. The $R^2$ for GenAI acceptance (0.65) showed a high and outstanding predictive ability,

suggesting the model's ability to anticipate teachers' acceptance of GenAI. According to Hair et al. [62], a $Q^2$ value of 0.35 or higher is deemed substantial. In this study, the $Q^2$ value for GenAI acceptance was found to be 0.38. This demonstrates the strong predictive relevance of the proposed model. Table 3 provides the detailed $R^2$ and $Q^2$ values.

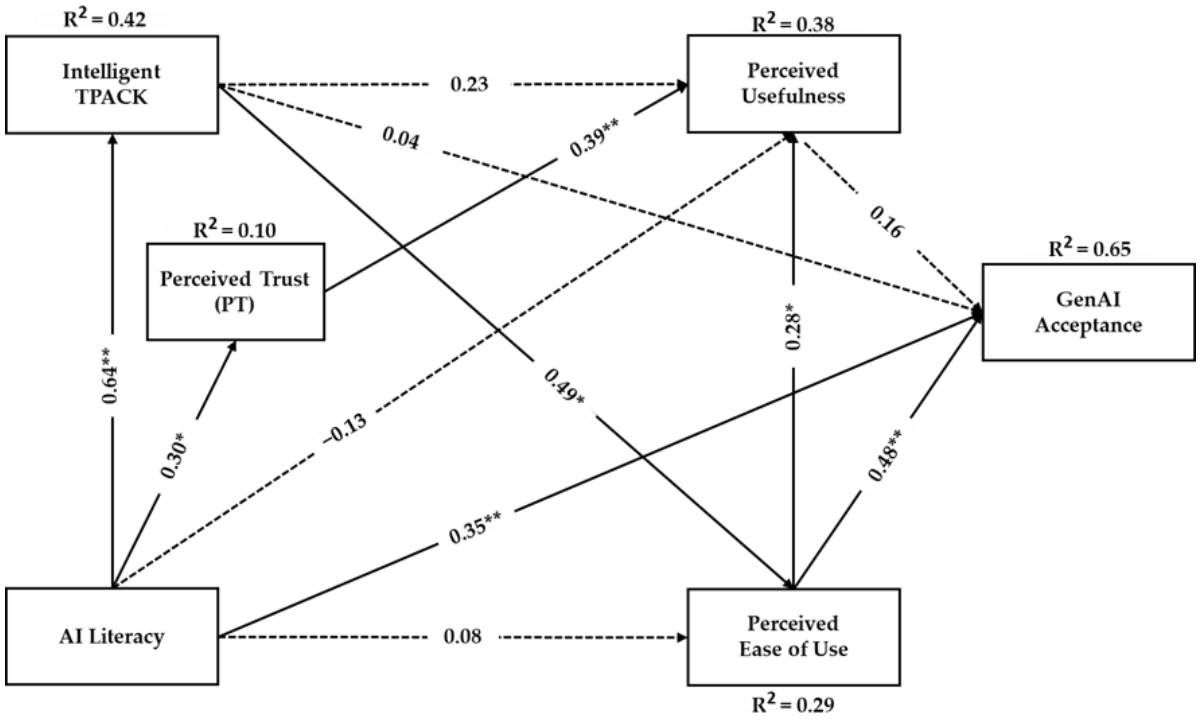


**Figure 2.** The results of standardized path coefficients (* *p*-value < 0.05, ** *p*-value < 0.001).

**Table 4.** Hypothesis test results.

| H | Direct Path → | | Effect β | SE | T-Value | Indirect Path → | 97% Confidence Interval Lower | Upper |
|---|---|---|---|---|---|---|---|---|
| H1 | Perceived usefulness → GenAI acceptance | Direct | 0.16 | 0.12 | 1.35 | | −0.01 | 0.43 |
| H2 | Perceived ease of use → GenAI acceptance | Direct | 0.48 | 0.10 | 4.53 ** | | 0.28 | 0.69 |
| H3 | Perceived ease of use → perceived usefulness | Direct | 0.28 | 0.12 | 2.23 * | | 0.04 | 0.53 |
| H4 | Intelligent TPACK → perceived usefulness | Direct | 0.23 | 0.15 | 1.47 | | −0.10 | 0.49 |
| H5 | Intelligent TPACK → perceived ease of use | Direct | 0.49 | 0.16 | 3.02 ** | | 0.17 | 0.80 |
| H6 | Intelligent TPACK → GenAI acceptance | Direct<br>In direct<br>Total | 0.04<br>0.23<br>0.33 | 0.09<br>0.09<br>0.10 | 0.40<br>2.36 *<br>3.20 ** | Intelligent TPACK → Perceived ease of use → GenAI acceptance | −0.16<br>0.08<br>0.13 | 0.19<br>0.46<br>0.53 |
| H7 | Al literacy → perceived usefulness | Direct<br>In direct<br>Total | −0.13<br>0.12<br>0.25 | 0.13<br>0.05<br>0.12 | 0.91<br>2.10 *<br>2.01 * | Al literacy → perceived trust → perceived usefulness | −0.39<br>0.02<br>0.02 | 0.15<br>0.23<br>0.49 |

**Table 4.** *Cont.*

| H | Direct Path → | | Effect β | SE | T-Value | Indirect Path → | 97% Confidence Interval Lower | Upper |
|---|---|---|---|---|---|---|---|---|
| H8 | Al literacy → perceived ease of use | Direct | 0.08 | 0.15 | 0.49 | Al literacy → intelligent TPACK → perceived ease of use | −0.21 | 0.37 |
| | | In direct | 0.31 | 0.12 | 2.54 * | | 0.10 | 0.56 |
| | | Total | 0.39 | 0.11 | 3.49 ** | | 0.18 | 0.61 |
| H9 | Al literacy → intelligent TPACK | Direct | 0.64 | 0.07 | 8.55 ** | | 0.48 | 0.77 |
| H10 | Al literacy → GenAI acceptance | Direct | 0.35 | 0.09 | 3.83 ** | | 0.13 | 0.50 |
| H11 | Al literacy → perceived trust | Direct | 0.30 | 0.12 | 2.44 * | | 0.06 | 0.54 |
| H12 | Perceived trust → perceived usefulness | Direct | 0.39 | 0.11 | 3.38 ** | | 0.16 | 0.60 |

β = path coefficient, SE = standard error, T-value is significant at ** $p < 0.001$, * $p < 0.05$.

## 7. Discussion and Implications

### *7.1. TAM Relationships*

The result for H1 showed a direct path coefficient (β = 0.16) with a non-significant t-value (1.35), indicating that perceived usefulness did not significantly affect GenAI acceptance in this study. Despite traditional models of TAM, which highly prioritize perceived usefulness [27], and unlike several studies in the field of AI technology acceptance [11,28], this hypothesis was not supported. Several factors might explain this weak relationship between perceived usefulness and GenAI acceptance. First, while perceived usefulness is a core factor in traditional technology adoption models, GenAI represents a novel and complex technology that educators may not yet fully understand or feel confident in using. Unlike more familiar technologies, such as learning management systems or digital tools that have clear, immediate benefits, the advanced capabilities of GenAI may appear distant or abstract in relation to educators' day-to-day needs. Second, the perceived complexity of GenAI may overshadow its usefulness. Educators may view it as a tool that requires significant learning and adaptation, with unclear short-term benefits. The perceived usefulness of GenAI in practice may diminish due to its perceived complexity and the absence of visible success stories or practical applications in educational settings. The perceived functional relevance of GenAI may decrease due to teachers' uncertainty about its effective integration into their existing workflows.

To better support the adoption of GenAI, professional development initiatives should focus on creating clearer, more context-specific demonstrations of GenAI's advantages, such as showcasing its ability to personalize learning or automate administrative tasks, thereby making the benefits more tangible and directly relevant to teachers' daily needs. Furthermore, this finding highlights the need for more targeted efforts to align GenAI technologies with the practical, everyday challenges faced by educators. Collaboration between GenAI developers and educators is key to creating tools that address specific classroom needs. For instance, GenAI developers could partner with educators to create tools that focus on automating specific teaching tasks, such as generating personalized learning plans or assisting in formative assessments by providing detailed, instant feedback based on student performance data. A potential application of this collaboration could involve designing GenAI-powered platforms that facilitate real-time classroom engagement, allowing teachers to adapt their instruction based on live feedback from students' interactions with digital content. Moreover, developers could co-create GenAI systems that support teachers in differentiating instruction for diverse student needs by recommending adaptive content or learning activities tailored to individual progress, thus addressing one of the core challenges teachers face in managing classroom diversity.

H2's result was significant (β = 0.48), strongly supporting the idea that perceived ease of use positively influences GenAI adoption. This aligns with TAM, where ease of use

is often found to be a critical factor in the acceptance and use of new technologies [27]. This means that if GenAI is perceived as easy to use, teachers are more likely to adopt it. This aligns with numerous studies suggesting that ease of use is a critical factor in AI technology adoption [9,29]. In the context of GenAI, complexity plays a unique role [1]. Unlike more conventional educational technologies, GenAI tools often incorporate advanced functionalities like natural language processing, content generation, and adaptive learning systems, potentially overwhelming or technically challenging for educators unfamiliar with these technologies. This highlights the importance of designing GenAI tools with simple, intuitive interfaces and providing comprehensive support systems to reduce complexity and encourage adoption. Reducing complexity through thoughtful design, simplified setup processes, and providing clear guidance on how GenAI can enhance teaching could mitigate concerns and improve adoption rates. Moreover, training and professional development should focus on demystifying the technology, reducing perceived complexity, and enhancing user confidence. Developers could integrate feedback mechanisms into these tools to refine and improve them based on real-world user experiences.

The support for H3 ($\beta = 0.28$) indicates a positive relationship between perceived ease of use and perceived usefulness. This suggests that when GenAI tools were easier to use, they were also perceived as more useful. This is consistent with suggestions in the literature that usability can enhance the value perceived by the user [2]. This could be due to the reduced cognitive load and increased accessibility of the technology, allowing educators to more readily integrate it into their pedagogical practices. For policymakers, efforts should be directed towards implementing professional development programs that specifically highlight features that save time or enhance lesson delivery, directly linking ease of use with practical benefits in the classroom. Clear guidelines on how GenAI reduces cognitive load and enhances accessibility could be included in national teacher training standards to ensure widespread adoption and effective use. Furthermore, GenAI developers should focus on designing tools that explicitly reduce cognitive load for educators and are easily accessible within classroom settings. Features that save time or streamline lesson delivery should be highlighted in user training materials and demonstrations. Developers could also collaborate with educators during the design process to identify which functions provide the greatest utility and ease of use, ensuring that their tools are practical for daily classroom application.

### *7.2. Intelligent TPACK Relationships*

H4 showed a non-significant result ($\beta = 0.23$), indicating that the direct impact of intelligent TPACK on perceived usefulness was not statistically significant. This suggests that simply possessing intelligent TPACK may not necessarily make GenAI tools seem more useful. This result is contrary to what was reported in previous studies [33,39,40], which emphasized the importance of TPACK in enhancing the perceived usefulness of technology. Several potential reasons could explain the lack of support for this hypothesis. One reason may be the context-specific nature of TPACK. The intelligent TPACK framework integrates knowledge from three key domains—technology, pedagogy, and content—but its application can vary depending on the type of technology being used [3]. GenAI, with its sophisticated functionalities such as natural language processing, content generation, and adaptive learning, may require a more specialized form of technological knowledge that goes beyond traditional TPACK. Educators may have a strong grasp of using conventional digital tools in a pedagogical context, but GenAI's complex, cutting-edge features could present a learning curve that renders existing TPACK frameworks less directly applicable. Thus, teachers may find themselves underprepared to fully integrate GenAI into their teaching despite their advanced TPACK knowledge.

Another possibility relates to technological familiarity, which likely differs among teachers. While some educators may be comfortable navigating various educational technologies, many may lack hands-on experience with GenAI tools specifically. This limited exposure can inhibit their ability to perceive the usefulness of GenAI in enhancing learning

outcomes, even if they have substantial expertise in blending technology with pedagogy and content. Furthermore, the rapid development of GenAI technologies may outpace the integration of such tools into professional development or teacher training programs. As a result, educators might not have had sufficient opportunities to explore or experiment with these tools in real-world educational settings. Without this practical experience, educators may struggle to see how GenAI can fit into their existing teaching frameworks, contributing to a disconnect between their knowledge and the perceived value of GenAI. This gap between theoretical understanding and practical application could point to a need for more targeted professional development. Existing TPACK-focused training may not sufficiently address the specific challenges posed by emerging technologies like GenAI. Therefore, professional development initiatives should incorporate hands-on experiences, case studies, and examples that showcase how GenAI can be effectively integrated into teaching. This would help educators bridge the gap between their TPACK knowledge and the specific applications of GenAI, making the technology seem more practical and useful in their teaching environments.

With a significant result ($\beta = 0.49$), H5 supports the idea that higher levels of intelligent TPACK improve the perceived ease of use of GenAI tools. This result is consistent with previous research on technology acceptance [41–43]. The support found for this hypothesis indicates that a solid grounding in TPACK can make GenAI tools appear easier for educators to use. Educational policymakers should advocate for and fund comprehensive TPACK-based training programs that specifically incorporate GenAI technologies, including GenAI. These programs should focus on hands-on, experiential learning, allowing educators to actively engage with GenAI tools. By integrating GenAI within the broader context of technological, pedagogical, and content knowledge, such training will equip teachers to seamlessly use GenAI in their classrooms. Policymakers can also develop incentives or standards for GenAI-focused teacher training to ensure nationwide consistency in GenAI adoption. In addition, developers should collaborate with educational institutions to create training modules that focus on practical, hands-on use of GenAI tools. These modules should align with the TPACK framework, enabling educators to see how GenAI integrates with their current pedagogical and technological practices.

The results for H6 showed that the direct influence of intelligent TPACK on GenAI acceptance was not significant ($\beta = 0.04$), suggesting that merely possessing intelligent TPACK does not directly lead to higher acceptance rates of GenAI technologies. This means that having the skills and knowledge to integrate technology into pedagogy is not enough on its own to ensure the adoption of new GenAI tools within educational settings [33]. One potential explanation is that GenAI tools involve unique functionalities, such as content generation and adaptive learning, which may not yet be fully addressed by current TPACK frameworks. As a result, even educators with strong TPACK skills may feel uncertain or hesitant about adopting GenAI, due to unfamiliarity with its specific applications or a lack of confidence in how it can fit within their pedagogical goals.

However, the indirect path through the perceived ease of use was significant (indirect $\beta = 0.23$), indicating that intelligent TPACK can facilitate GenAI adoption by making the technology easier to use. This suggests that while intelligent TPACK may not directly motivate teachers to accept GenAI tools, as previous studies have indicated in technology acceptance [40,42,43], it plays a crucial role in reducing the complexity or improving the usability of these tools, which in turn influences adoption decisions. The total effect (total $\beta = 0.33$) also highlighted the overall impact when both direct and indirect paths were considered. Developers should work closely with educators to ensure that their GenAI tools are designed with user-friendliness as a priority. This involves creating intuitive interfaces and providing clear, practical training resources that demonstrate how GenAI can be seamlessly incorporated into existing pedagogical frameworks like TPACK. Developers could also offer support tools and tutorials that directly link TPACK competence with the day-to-day use of GenAI, thereby reducing barriers to adoption and making the technology more accessible to educators.

### 7.3. AI Literacy Relationships

The results for H7 presented an interesting dynamic, where the direct relationship between AI literacy and perceived usefulness was not significant ($\beta = -0.13$), suggesting that on its own, AI literacy does not directly lead to teachers viewing GenAI as more useful. This could indicate that simply understanding AI concepts and being literate in AI technology is not enough to influence educators' perceptions of how useful GenAI tools are in their teaching practices. However, the significant indirect path through perceived trust (indirect $\beta = 0.12$) reveals an important mediating role of trust in the technology. This suggests that AI literacy alone may not change perceptions of usefulness unless it is accompanied by a sense of trust in the technology. Educators need to feel confident that GenAI is reliable, transparent, and beneficial before they can see it as useful in their educational context. The total impact (total $\beta = 0.25$) demonstrates that while AI literacy does not directly drive perceived usefulness, its indirect effects—mediated by perceived trust—combine to create a positive overall perception of GenAI's utility.

This indicates that AI literacy indirectly influences perceived usefulness by enhancing trust in GenAI technologies. Trust acts as a validation mechanism [58]. An AI-literate teacher might understand the potential of GenAI, but it is their trust in the tool's consistent performance that will determine if they perceive it as genuinely useful [5,56]. The implication is that educational programs should focus not only on improving AI literacy but also on building trust in GenAI technologies through transparency, reliability, and demonstrated effectiveness. Enhancing AI literacy should therefore include elements of critical thinking about technology, ethical considerations, and understanding GenAI's capabilities and limitations, which could foster a more trusting and thus more receptive attitude towards GenAI.

The results for H8 also presented an interesting dynamic. The direct effect of AI literacy on perceived ease of use was not significant ($\beta = 0.08$), suggesting that more knowledge about GenAI does not straightforwardly make GenAI tools easier to use. This might be because the technical complexity of GenAI can still pose usability challenges, regardless of one's understanding of the underlying concepts. One possible explanation for this result is the inherent technical complexity of GenAI tools. Even educators with a solid grasp of AI concepts may still find these technologies challenging to navigate, particularly if the tools require advanced technical skills or involve complex functionalities like natural language processing or content generation.

However, the significant indirect path through intelligent TPACK (indirect $\beta = 0.31$) implies that AI literacy helped improve the perceived ease of use by enhancing intelligent TPACK. This can be interpreted to mean that AI literacy enriches teachers' ability to integrate GenAI into their pedagogical practice effectively, thus making GenAI tools seem easier to use [4,35,48]. This highlights the interconnectedness of AI literacy and intelligent TPACK skills, suggesting that efforts to improve ease of use should focus on strengthening both AI knowledge and its pedagogical integration. The total effect being significant (total $\beta = 0.39$) further highlights the substantial overall impact of AI literacy on perceived ease of use when mediated through enhanced TPACK capabilities. Overall, these finding underscore the need for more hands-on training and practical experience with GenAI so that educators can build familiarity with the tools in action, rather than relying solely on theoretical knowledge.

The strong support for H9 ($\beta = 0.64$) indicates a direct and robust relationship between AI literacy and intelligent TPACK. The support for this hypothesis underscores the foundational role of AI literacy in enhancing educators' intelligent TPACK. This suggests that AI literacy is a key component in equipping teachers with the necessary skills and knowledge to integrate GenAI effectively into their teaching [45,48]. The implication is that educational policies should prioritize AI literacy as a core component of teacher education curricula and training programs to prepare educators for the evolving technological landscape, focusing on both the theoretical and practical aspects of GenAI [4,31]. Policymakers can implement initiatives that integrate AI literacy into teacher certification requirements, ensuring that

educators are equipped with the knowledge and skills to effectively utilize GenAI in their classrooms [64]. Additionally, targeted funding should be allocated to support ongoing AI literacy training and development, ensuring continuous growth and adaptation to new technological advancements. Additionally, developers can contribute to this effort by creating educational resources and training modules that specifically address AI literacy in both theoretical and practical terms. These resources should be designed to complement teacher education programs, providing real-world examples and hands-on practice with GenAI tools. Developers can also collaborate with educational institutions to ensure that their tools are accessible and align with the learning objectives of AI literacy curricula, thus supporting educators in mastering the application of GenAI in their teaching.

H10 was significantly supported (β = 0.35), showing that AI literacy directly influenced GenAI acceptance. AI literacy improves understanding of GenAI's advantages and functions, which encourages a favorable view of its acceptance and use in educational contexts [3,47,48,64]. Like H9, this result accentuates the need for comprehensive AI education programs that address both theoretical knowledge and practical applications, promoting a more informed and positive attitude towards GenAI. These programs could thus focus on comprehensive GenAI education that goes beyond basic functionality to include ethical usage, potential biases in GenAI, and strategies for integrating GenAI tools into pedagogical practices.

### *7.4. Perceived Trust Relationships*

With a significant result (β = 0.30), H11 was supported, confirming that AI literacy enhances perceived trust in GenAI tools. This finding underscores the importance of increasing AI literacy to build trust in GenAI technologies, which is likely to influence their broader acceptance and use [50,59]. However, to further strengthen trust, efforts to enhance AI literacy should also address critical issues such as privacy, bias, and accountability, which are central to ethical concerns around GenAI. These factors can significantly affect trust, as educators need to be assured that GenAI tools operate securely, fairly, and transparently. There is a clear need for transparency about how GenAI operates, including its decision-making processes, data handling practices, and limitations. Educators must feel confident in their understanding of these systems, not only in terms of functionality but also regarding their ethical implications. Building this trust can be achieved through open forums, detailed documentation, and ethical training that emphasizes privacy protection, fairness, and accountability in GenAI use.

Similarly, H12 was strongly supported (β = 0.39), indicating that greater trust in GenAI enhances its perceived usefulness. This aligns with the idea that trust is a key factor in determining how useful technology is perceived to be [5,56,58]. The implication is that building trust in GenAI tools through demonstrated reliability, effectiveness, and strong ethical practices can significantly enhance their perceived usefulness, making them a more attractive option for educators. This includes addressing concerns related to bias and accountability, which can affect educators' confidence in GenAI's ability to serve their needs equitably and transparently. Ongoing dialogue between GenAI developers and educators is crucial, and should be supported by policy frameworks that ensure these technologies are not only reliable and effective but also equitable, secure, and aligned with educational goals.

## 8. Conclusions, Limitations, and Future Work

This study explored the factors influencing the acceptance of GenAI technologies among educators in higher-education settings. It was driven by a framework integrating concepts from TAM, intelligent TPACK, AI literacy, and perceived trust to evaluate their impacts on university teachers' acceptance of GenAI technologies. The results revealed a complex landscape in which perceived ease of use significantly influenced both the acceptance of GenAI and its perceived usefulness, underscoring the importance of user-friendly design in GenAI technologies. Contrary to expectations, perceived usefulness

did not have a direct impact on acceptance, indicating potential gaps in aligning GenAI's capabilities with educators' needs. AI literacy emerged as a crucial factor, directly affecting acceptance, intelligent TPACK, and perceived trust in GenAI.

However, the anticipated direct impacts of intelligent TPACK on the perceived usefulness and acceptance of GenAI were not supported. This finding was particularly interesting, as it suggests that having a robust TPACK framework does not automatically translate into positive perceptions or acceptance of GenAI. Instead, the relationship between intelligent TPACK and GenAI acceptance was mediated by perceived ease of use, indicating that the usability of GenAI tools might be a more critical factor than previously thought. Perceived trust mediated the impact of AI literacy on teachers' perceived usefulness of GenAI. This suggests that while AI literacy may provide educators with foundational knowledge of GenAI, it is their perceived trust that transforms this knowledge into confidence in practical application.

This study, however, is not without limitations. This study extends the technology acceptance model (TAM) by incorporating constructs such as AI literacy, intelligent TPACK, and perceived trust. It is important to acknowledge that the research primarily focused on individual-level factors influencing the acceptance of GenAI. Future research could expand on this work by exploring how organizational factors, cognitive load, and pedagogical integration challenges influence the adoption of AI technologies, thereby offering a more comprehensive understanding of the multifaceted barriers and enablers in educational contexts. In addition, the reliance on self-reported measures may have introduced bias, and such measures do not capture the real-time use or integration of GenAI in educational practices. Future studies could incorporate observational or charted data that provide more objective insights into how GenAI is actually used in educational settings, rather than relying solely on educator self-reports.

Furthermore, the cross-sectional nature of the study limited its ability to capture the dynamic process of technology adoption over time. Educators' attitudes towards and proficiency with GenAI may evolve as they gain more experience and as the technology itself matures. Longitudinal research could track these changes over time, providing insights into the evolution of acceptance and the long-term effects of GenAI integration in educational practices. While this study offers valuable insights into GenAI adoption among educators, it primarily focuses on a specific educational context. As a result, cultural and regional differences, which may play a significant role in GenAI adoption, were not within the scope of this research. By narrowing the focus, we were able to provide a detailed examination of key factors such as AI literacy, intelligent TPACK, and perceived trust. However, future research could build on these findings by exploring how cultural and regional variations influence educators' acceptance of GenAI, offering a more comprehensive understanding across diverse global contexts.

**Funding:** The Deanship of Scientific Research, Vice Presidency of Graduate Studies and scientific research at King Faisal University (grant A374).

**Institutional Review Board Statement:** Research Ethics Committee at King Faisal University, KFU-REC-2022-OCT-ETHICS258, issued on 25 October 2022.

**Informed Consent Statement:** Informed consent was obtained from all subjects involved in the study.

**Data Availability Statement:** The datasets generated during and/or analyzed during the current study are not publicly available due to privacy issues.

**Conflicts of Interest:** The author declares no conflicts of interest.

## Appendix A

| Survey |
| --- |
| **Construct (items)** |
| ***Perceived ease of use*** |
| 1. Learning how to use GenAI tools is easy for me. |
| 2. My interaction with GenAI tools is clear and simple. |
| 3. I find GenAI tools easy to use for my teaching. |
| 4. It is easy for me to become skillful in using GenAI tools. |
| ***Perceived usefulness*** |
| 1. I find GenAI tools useful for performing my teaching duties. |
| 2. Using GenAI tools increases my chances of achieving high job performance. |
| 3. Using GenAI tools helps me accomplish my teaching tasks effortlessly. |
| 4. Using GenAI tools increases my productivity. |
| ***Perceived trust*** |
| 1. I will use the GenAI tools if I feel that the content is trustworthy. |
| 2. I will use the GenAI tools if I feel that these tools provide reliable information. |
| 3. I will use the GenAI tools if I feel that these tools meet my expectations. |
| 4. I will use the GenAI tools if I feel that these tools are secure. |
| 5. I will use the GenAI tools if the data use poses no risks to teacher and learner privacy. |
| ***AI literacy*** |
| 1. I can use AI applications or products to improve my work efficiency. |
| 2. I can evaluate the capabilities and limitations of an AI application or product after using it for a while. |
| 3. I always comply with ethical principles when using AI applications or products. |
| 4. I am alert to privacy and information security issues when using AI applications or products. |
| ***Intelligent TPACK*** |
| 1. In teaching my field, I know how to use different GenAI tools for adaptive and real-time feedback. |
| 2. In teaching my field, I know how to use different GenAI tools for personalized learning. |
| 3. In teaching my field, I know how to use different GenAI tools for generate quiz and assessments. |
| 4. I can teach lessons that appropriately combine my teaching content, GenAI tools, and teaching strategies. |
| 5. I can take a leadership role among my colleagues in the integration of GenAI tools into our teaching field. |
| 6. I can select various GenAI tools to monitor students' learning in my teaching process. |
| ***GenAI acceptance*** |
| 1. I look forward to use GenAI tools in my teaching. |
| 2. I intend to use GenAI tools in my future teaching. |
| 3. I plan to use GenAI tools in my future teaching. |
| 4. I support the adoption of GenAI tools in higher education. |

## References


1. Lodge, J.M.; Thompson, K.; Corrin, L. Mapping out a research agenda for generative artificial intelligence in tertiary education. *Australas. J. Educ. Technol.* **2023**, *39*, 1–8. [CrossRef]
2. Davis, F.D. A technology Acceptance Model for Empirically Testing New End-User Information Systems: Theory and Results. Master's Thesis, Massachusetts Institute of Technology, Cambridge, MA, USA, 1985.
3. Celik, I. Towards intelligent-TPACK: An empirical study on teachers' professional knowledge to ethically integrate artificial intelligence (AI)-based tools into education. *Comput. Hum. Behav.* **2023**, *138*, 107468. [CrossRef]
4. Laupichler, M.C.; Aster, A.; Schirch, J.; Raupach, T. Artificial intelligence literacy in higher and adult education: A scoping literature review. *Comput. Educ. Artif. Intell.* **2022**, *3*, 100101. [CrossRef]
5. Nazaretsky, T.; Ariely, M.; Cukurova, M.; Alexandron, G. Teachers' trust in AI-powered educational technology and a professional development program to improve it. *Br. J. Educ. Technol.* **2022**, *53*, 914–931. [CrossRef]
6. Crompton, H.; Burke, D. Artificial intelligence in higher education: The state of the field. *Int. J. Educ. Technol. High. Educ.* **2023**, *20*, 1–22. [CrossRef]
7. Bozkurt, A. Generative artificial intelligence (AI) powered conversational educational agents: The inevitable paradigm shift. *Asian J. Distance Educ.* **2023**, *18*, 198–204.
8. Mujiono, M. Educational collaboration: Teachers and artificial intelligence. *J. Kependidikan J. Has. Penelit. Dan Kaji. Kepustakaan Di Bid. Pendidik. Pengajaran Dan Pembelajaran* **2023**, *9*, 618–632. [CrossRef]
9. Kaplan-Rakowski, R.; Grotewold, K.; Hartwick, P.; Papin, K. Generative AI and teachers' perspectives on its implementation in education. *J. Interact. Learn. Res.* **2023**, *34*, 313–338.
10. Al-Amri, N.A.; Al-Abdullatif, A.M. Drivers of Chatbot Adoption among K–12 Teachers in Saudi Arabia. *Educ. Sci.* **2024**, *14*, 1034. [CrossRef]
11. Chen, X.; Xie, H.; Zou, D.; Hwang, G.J. Application and theory gaps during the rise of artificial intelligence in education. *Comput. Educ. Artif. Intell.* **2020**, *1*, 100002. [CrossRef]
12. Zhang, K.; Aslan, A.B. AI technologies for education: Recent research & future directions. *Comput. Educ. Artif. Intell.* **2021**, *2*, 100025. [CrossRef]

13. Baidoo-Anu, D.; Ansah, L.O. Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *J. AI* **2023**, *7*, 52–62. [CrossRef]
14. Lo, C.K. What is the impact of ChatGPT on education? A rapid review of the literature. *Educ. Sci.* **2023**, *13*, 410. [CrossRef]
15. Ge, J.; Lai, J.C. Artificial intelligence-based text generators in hepatology: ChatGPT is just the beginning. *Hepatol. Commun.* **2023**, *7*, e0097. [CrossRef] [PubMed]
16. Rane, N. Enhancing the Quality of Teaching and Learning through ChatGPT and Similar Large Language Models: Challenges, Future Prospects, and Ethical Considerations in Education. Available online: https://ssrn.com/abstract=4599104 (accessed on 20 October 2024).
17. Al-Abdullatif, A.M.; Al-Dokhny, A.A.; Darwish, A.M. Implementing the Bashayer chatbot in Saudi higher education: Measuring the influence on students' motivation and learning strategies. *Front. Psychol.* **2023**, *14*, 1129070. [CrossRef]
18. Grassini, S. Shaping the future of education: Exploring the potential and consequences of AI and ChatGPT in educational settings. *Educ. Sci.* **2023**, *13*, 692. [CrossRef]
19. Elmohandes, N.; Marghany, M. Effective or ineffective? Using ChatGPT for staffing in the hospitality industry. *Eur. J. Tour. Res.* **2024**, *36*, 3617. [CrossRef]
20. Gill, S.S.; Xu, M.; Patros, P.; Wu, H.; Kaur, R.; Kaur, K.; Fuller, S.; Singh, M.; Arora, P.; Parlikad, A.K.; et al. Transformative effects of ChatGPT on modern education: Emerging era of AI chatbots. *Internet Things Cyber-Phys. Syst.* **2024**, *4*, 19–23. [CrossRef]
21. Al-Abdullatif, A.M. Modeling students' perceptions of chatbots in learning: Integrating technology acceptance with the value-based adoption model. *Educ. Sci.* **2023**, *13*, 1151. [CrossRef]
22. Wach, K.; Duong, C.D.; Ejdys, J.; Kazlauskaitė, R.; Korzynski, P.; Mazurek, G.; Paliszkiewicz, J.; Ziemba, E.W. The dark side of generative artificial intelligence: A critical analysis of controversies and risks of ChatGPT. *Entrep. Bus. Econ. Rev.* **2023**, *11*, 7–30. [CrossRef]
23. Chan, C.K.Y.; Lee, K.K. The AI generation gap: Are Gen Z students more interested in adopting generative AI such as ChatGPT in teaching and learning than their Gen X and millennial generation teachers? *Smart Learn. Environ.* **2023**, *10*, 60. [CrossRef]
24. García-Peñalvo, F.J.; Llorens Largo, F.; Vidal, J. The new reality of education in the face of advances in generative artificial intelligence. *Ibero-Am. J. Distance Educ.* **2023**, *27*, 9–39. [CrossRef]
25. Tossell, C.C.; Tenhundfeld, N.L.; Momen, A.; Cooley, K.; De Visser, E.J. Student perceptions of ChatGPT use in a college essay assignment: Implications for learning, grading and trust in artificial intelligence. *IEEE Trans. Learn. Technol.* **2024**, *17*, 1069–1081. [CrossRef]
26. Yu, H.; Guo, Y. Generative artificial intelligence empowers educational reform: Current status, issues, and prospects. *Front. Educ.* **2023**, *8*, 1183162. [CrossRef]
27. Davis, F.D. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Q.* **1989**, *13*, 319–340. [CrossRef]
28. Liao, Y.K.; Wu, W.Y.; Le, T.Q.; Phung, T.T.T. The integration of the technology acceptance model and value-based adoption model to study the adoption of e-learning: The moderating role of e-WOM. *Sustainability* **2022**, *14*, 815. [CrossRef]
29. Malik, R.; Shrama, A.; Trivedi, S.; Mishra, R. Adoption of chatbots for learning among university students: Role of perceived convenience and enhanced performance. *Int. J. Emerg. Technol. Learn.* **2021**, *16*, 200–212. [CrossRef]
30. Rosli, M.S.; Saleh, N.S.; Ali, A.M.; Abu Bakar, S.; Tahir, L.M. A systematic review of the technology acceptance model for the sustainability of higher education during the COVID-19 pandemic and identified research gaps. *Sustainability* **2022**, *14*, 11389. [CrossRef]
31. Ng, D.T.K.; Leung, J.K.L.; Chu, S.K.W.; Qiao, M.S. Conceptualizing AI literacy: An exploratory review. *Comput. Educ. Artif. Intell.* **2021**, *2*, 100041. [CrossRef]
32. Mishra, P.; Koehler, M.J. Technological pedagogical content knowledge: A framework for teacher knowledge. *Teach. Coll. Rec.* **2006**, *108*, 1017–1054. [CrossRef]
33. Joo, Y.J.; Park, S.; Lim, E. Factors influencing preservice teachers' intention to use technology: TPACK, teacher self-efficacy, and technology acceptance model. *J. Educ. Technol. Soc.* **2018**, *21*, 48–59. Available online: http://www.jstor.org/stable/26458506 (accessed on 20 October 2024).
34. Kim, S.; Jang, Y.; Choi, S.; Kim, W.; Jung, H.; Kim, S.; Kim, H. Analyzing teacher competency with TPACK for K-12 AI education. *KI—Künstliche Intell.* **2021**, *35*, 139–151. [CrossRef]
35. Velander, J.; Taiye, M.A.; Otero, N.; Milrad, M. Artificial intelligence in K-12 education: Eliciting and reflecting on Swedish teachers' understanding of AI and its implications for teaching & learning. *Educ. Inf. Technol.* **2023**, *29*, 4085–4105. [CrossRef]
36. Mishra, P.; Warr, M.; Islam, R. TPACK in the age of ChatGPT and Generative AI. *J. Digit. Learn. Teach. Educ.* **2023**, *39*, 235–251. [CrossRef]
37. Zawacki-Richter, O.; Marín, V.I.; Bond, M.; Gouverneur, F. Systematic review of research on artificial intelligence applications in higher education: Where are the educators? *Int. J. Educ. Technol. High. Educ.* **2019**, *16*, 39. [CrossRef]
38. Scherer, R.; Siddiq, F.; Tondeur, J. The technology acceptance model (TAM): A meta-analytic structural equation modeling approach to explaining teachers' adoption of digital technology in education. *Comput. Educ.* **2019**, *128*, 13–35. [CrossRef]
39. Hsu, L. Examining EFL teachers' technological pedagogical content knowledge and the adoption of mobile-assisted language learning: A partial least square approach. *Comput. Assist. Lang. Learn.* **2016**, *29*, 1287–1297. [CrossRef]

40. Yang, J.; Wang, Q.; Wang, J.; Huang, M.; Ma, Y. A study of K-12 teachers' TPACK on the technology acceptance of E-schoolbag. *Interact. Learn. Environ.* **2021**, *29*, 1062–1075. [CrossRef]
41. Mailizar, M.; Hidayat, M.; Al-Manthari, A. Examining the impact of mathematics teachers' TPACK on their acceptance of online professional development. *J. Digit. Learn. Teach. Educ.* **2021**, *37*, 196–212. [CrossRef]
42. Jang, J.; Ko, Y.; Shin, W.S.; Han, I. Augmented reality and virtual reality for learning: An examination using an extended technology acceptance model. *IEEE Access* **2021**, *9*, 6798–6809. [CrossRef]
43. Thohir, M.A.; Ahdhianto, E.; Mas'ula, S.; Yanti, F.A.; Sukarelawan, M.I. The effects of TPACK and facility condition on preservice teachers' acceptance of virtual reality in science education course. *Contemp. Educ. Technol.* **2023**, *15*, ep407. [CrossRef] [PubMed]
44. Kong, S.C.; Cheung, W.M.Y.; Zhang, G. Evaluation of an artificial intelligence literacy course for university students with diverse study backgrounds. *Comput. Educ. Artif. Intell.* **2021**, *2*, 100026. [CrossRef]
45. Zhao, L.; Wu, X.; Luo, H. Developing AI literacy for primary and middle school teachers in China: Based on a structural equation modelling analysis. *Sustainability* **2022**, *14*, 14549. [CrossRef]
46. Long, D.; Magerko, B. What is AI literacy? Competencies and design considerations. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, Honolulu HI, USA, 23 April 2020; pp. 1–16. [CrossRef]
47. Wang, B.; Rau, P.L.P.; Yuan, T. Measuring user competence in using artificial intelligence: Validity and reliability of artificial intelligence literacy scale. *Behav. Inf. Technol.* **2023**, *42*, 1324–1337. [CrossRef]
48. Kong, S.C.; Cheung, W.M.Y.; Zhang, G. Evaluating artificial intelligence literacy courses for fostering conceptual learning, literacy and empowerment in university students: Refocusing to conceptual building. *Comput. Hum. Behav. Rep.* **2022**, *7*, 100223. [CrossRef]
49. Su, J.; Yang, W. Artificial intelligence (AI) literacy in early childhood education: An intervention study in Hong Kong. *Interact. Learn. Environ.* **2023**, 1–15. [CrossRef]
50. Lee, T.; Lee, B.K.; Lee-Geiller, S. The effects of information literacy on trust in government websites: Evidence from an online experiment. *Int. J. Inf. Manag.* **2020**, *52*, 102098. [CrossRef]
51. Uygun, D. Teachers' perspectives on artificial intelligence in education. *Adv. Mob. Learn. Educ. Res.* **2024**, *4*, 931–939. [CrossRef]
52. Yue, M.; Jong, M.S.Y.; Dai, Y. Pedagogical design of K-12 artificial intelligence education: A systematic review. *Sustainability* **2022**, *14*, 15620. [CrossRef]
53. Chatterjee, S.; Bhattacharjee, K.K. Adoption of artificial intelligence in higher education: A quantitative analysis using structural equation modelling. *Educ. Inf. Technol.* **2020**, *25*, 3443–3463. [CrossRef]
54. Al Darayseh, A. Acceptance of artificial intelligence in teaching science: Science teachers' perspective. *Comput. Educ. Artif. Intell.* **2023**, *4*, 100132. [CrossRef]
55. Mou, J.; Shin, D.H.; Cohen, J. Understanding trust and perceived usefulness in the consumer acceptance of an e-service: A longitudinal investigation. *Behav. Inf. Technol.* **2017**, *36*, 125–139. [CrossRef]
56. Amoozadeh, M.; Daniels, D.; Nam, D.; Chen, S.; Hilton, M.; Ragavan, S.S.; Alipour, M.A. Trust in generative AI among students: An exploratory study. In Proceedings of the 55th ACM Technical Symposium on Computer Science Education, Portland, OR, USA, 7 March 2024; pp. 67–73. [CrossRef]
57. Baek, T.H.; Kim, M. Is ChatGPT scary good? How user motivations affect creepiness and trust in generative artificial intelligence. *Telemat. Inform.* **2023**, *83*, 102030. [CrossRef]
58. Gillath, O.; Ai, T.; Branicky, M.S.; Keshmiri, S.; Davison, R.B.; Spaulding, R. Attachment and trust in artificial intelligence. *Comput. Hum. Behav.* **2021**, *115*, 106607. [CrossRef]
59. Nazaretsky, T.; Cukurova, M.; Alexandron, G. An instrument for measuring teachers' trust in AI-based educational technology. In Proceedings of the LAK22: 12th International Learning Analytics and Knowledge Conference, Online, 21 March 2022; pp. 56–66. [CrossRef]
60. Nazzal, A.; Thoyib, A.; Zain, D.; Hussein, A.S. The effect of digital literacy and website quality on purchase intention in internet shopping through mediating variable: The case of internet users in Palestine. *Webology* **2022**, *19*, 2414–2434. [CrossRef]
61. Kline, R.B. *Principles and Practice of Structural Equation Modeling*, 3rd ed.; The Guilford Press: New York, NY, USA, 2015.
62. Hair, J.F.; Risher, J.J.; Sarstedt, M.; Ringle, C.M. When to use and how to report the results of PLS-SEM. *Eur. Bus. Rev.* **2019**, *31*, 2–24. [CrossRef]
63. Henseler, J.; Ringle, C.M.; Sarstedt, M. A new criterion for assessing discriminant validity in variance-based structural equation modeling. *J. Acad. Mark. Sci.* **2015**, *43*, 115–135. [CrossRef]
64. Al-Abdullatif, A.M.; Alsubaie, M.A. ChatGPT in Learning: Assessing Students' Use Intentions through the Lens of Perceived Value and the Influence of AI Literacy. *Behav. Sci.* **2024**, *14*, 845. [CrossRef]


**Disclaimer/Publisher's Note:** The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.